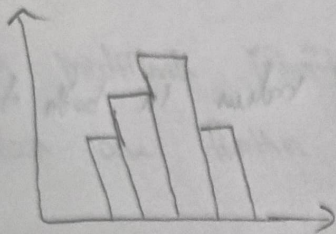


Data visualization

Charts:

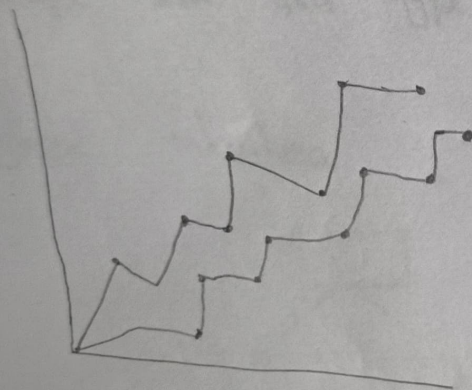
Bar Chart:



Types of data: Categorical, quantitative

- * Bar chart use data to compare data across categories.
- * Avoid bar chart: When there are too many categories or data is continuous.

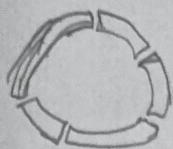
Line Chart:



Types of data: Continuous, time series

- * Data connected by lines. X-axis represents time, and Y-axis represents value.
- Avoid: use line chart when there is large order, relationship between data.

Donut chart:

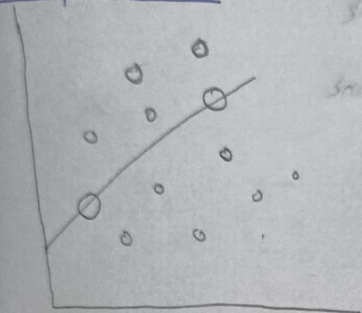


Types of data: Categorical, proportional

When to use it: To show proportional data

Avoid: Many categories of data

Scatterplot: Chart:



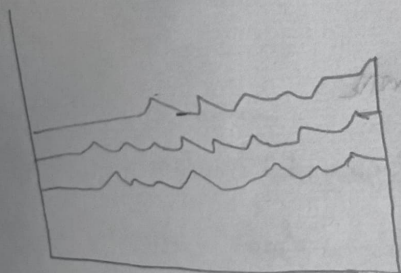
Types of data: Continuous, bivariate

Usage: Scatterplot displays relationship b/w two variables

Shows: plots two variables one on X axis and another Y axis

Avoid: When data is irrelevant to each other

Area Chart:



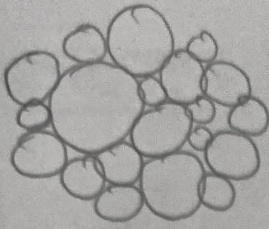
Types of data: Continuous, time series

Usage: To show volume or magnitude of data over time

Shows: Similar to line chart but area b/w lines and X axis is filled

Avoid: Multiple data with same value leads to overlap.

Bubble Chart:



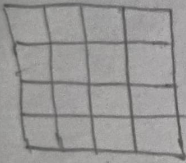
Type of data: Continuous, multivariate

Usage: to display multi relation data

Shows: bubble chart is variety of scatterplot, size of bubble represents data value

Avoid: when data is same value

Heatmap:



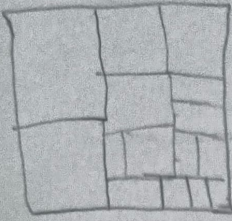
Type of data: continuous, multivariate

Usage: display relationship b/w two variables in heat intensity

Shows: Heatmap is color map to display value of each cell in matrix. One variable on x axis other on y axis. Dark color represents higher value, light color represents lower value.

Avoid: when data is irrelevant

Tree map:



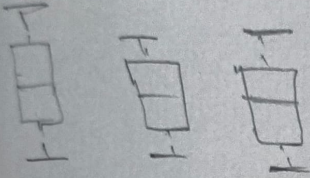
Type of data: Categorical, hierarchical

usage: To display hierarchical data or to show proportion of each category

Shows: Data in hierarchical order of boxes

Avoids: .

Box plot:

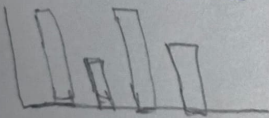


Type of data: Continuous, univariate

usage: to display distribution of data and detect outliers.

Shows: A rectangular plot to represent interquartile range (IQR) and whiskers to show range of data.

Stacked Bar chart:



Type of data: Categorical, quantitative

usage: A stacked bar chart to compare data across categories and shows composition of each category

Shows: A stacked chart with length of each segment represents value.